# Estimated Annual Agricultural Pesticide Use
# Yakima County 2012-2016

21 July 2026

## Summary

Table 1 lists the rank and best available estimates of mass applied (kg) for the 80 most common agricultural pesticides in Yakima County during 2016, as reported by USGS Estimated Annual Agricultural Pesticide Use data. The table also includes rank by 3-year average (2014-2016) and 5-year average (2012-2016) estimates of mass applied. Rankings and estimates are based on the *EPest-high method* described by Theilen and Stone (2013) and Baker and Stone (2015):

> "EPest-low and EPest-high, are used to estimate a range of pesticide use. Both EPest-low and EPest-high methods incorporate proprietary surveyed rates for Crop Reporting Districts (CRDs), but EPest-low and EPest-high estimates differ in how they treat situations when a CRD was surveyed and pesticide use was not reported for a particular crop present in the CRD. In these situations, EPest-low assumes zero use in the CRD for that pesticide-by-crop combination. EPest-high, however, treats the unreported use for that pesticide-by- crop combination in the CRD as missing data. In this case, pesticide-by-crop use rates from neighboring CRDs or CRDs within the same region are used to estimate the pesticide-by-crop EPest-high rate for the CRD." *-Pesticide National Synthesis Project webpage*

USGS lists several caveats of these data, in particular:

- Data for 2013-2016 are preliminary (i.e. "using projected county crop acres from the previous Census of Agriculture and are expected to be revised upon availability of updated crop acreages in the following Census of Agriculture")
- Reliability decreases with scale (e.g. "detailed interpretation of where and how much use occurs within a county is not appropriate")
- EPest-low estimates are more likely to reflect state-based restrictions on pesticide use than EPest-high estimates

For a more detailed look at the range (EPest-low and EPest-high) of 1-yr, 3-yr, and 5-yr data, please see Tables 2, 3, and 4, respectively.

Table 1: Top 80 EPest-high Estimates for Yakima County, 1-Yr (2016), 3-Yr Avg (2014-2016), 5-Yr Avg (2012-2016)

| COMPOUND | RANK_1YR | RANK_3YR | RANK_5YR | EPEST_HIGH_KG_1YR_ONLY | EPEST_HIGH_KG_3YR_AVG | EPEST_HIGH_KG_5YR_AVG |
|---|---|---|---|---|---|---|
| PETROLEUM OIL | 1 | 1 | 1 | 1715390 | 1523056 | 1431482 |
| SULFUR | 2 | 2 | 2 | 397630 | 341576 | 285379 |
| CALCIUM POLYSULFIDE | 3 | 3 | 3 | 197137 | 199437 | 164121 |
| KAOLIN CLAY | 4 | 4 | 4 | 153162 | 127777 | 131143 |
| GLYPHOSATE | 5 | 6 | 6 | 89873 | 64993 | 56610 |
| COPPER | 6 | 8 | 11 | 51880 | 38083 | 28067 |
| CHLORPYRIFOS | 7 | 7 | 9 | 40284 | 43552 | 40193 |
| COPPER SULFATE | 8 | 13 | 17 | 35194 | 19198 | 15423 |
| PARAQUAT | 9 | 16 | 15 | 26162 | 16855 | 16586 |
| METAM | 10 | 10 | 8 | 23355 | 33961 | 40436 |
| DICHLOROPROPENE | 11 | 9 | 7 | 21475 | 36697 | 41146 |
| 2,4-D | 12 | 12 | 13 | 18582 | 21719 | 21304 |
| SULFURIC ACID | 13 | 17 | 14 | 17168 | 15226 | 18163 |
| CARBARYL | 14 | 11 | 12 | 14195 | 24652 | 21888 |
| PENDIMETHALIN | 15 | 20 | 16 | 12959 | 14692 | 15812 |
| ZIRAM | 16 | 21 | 24 | 11924 | 11906 | 9892 |
| METAM POTASSIUM | 17 | 18 | 10 | 10763 | 15213 | 29218 |
| MANCOZEB | 18 | 19 | 19 | 10218 | 14701 | 13483 |
| ATRAZINE | 19 | 39 | 40 | 7221 | 3660 | 4056 |
| TRIFLUMIZOLE | 20 | 22 | 28 | 7031 | 8596 | 7915 |
| POTASSIUM BICARBONATE | 21 | 26 | 29 | 6593 | 7877 | 6942 |
| BUPROFEZIN | 22 | 43 | 54 | 5292 | 3267 | 2503 |
| GLUFOSINATE | 23 | 34 | 26 | 5018 | 4269 | 8831 |
| TRICLOPYR | 24 | 44 | 18 | 4962 | 3118 | 14269 |
| BOSCALID | 25 | 32 | 39 | 4376 | 4467 | 4099 |
| BACILLUS THURINGIENSIS | 26 | 47 | 53 | 4314 | 3014 | 2577 |
| PHOSMET | 27 | 28 | 33 | 3909 | 6154 | 5792 |
| MALATHION | 28 | 50 | 56 | 3888 | 2710 | 2407 |
| DIURON | 29 | 29 | 36 | 3519 | 5254 | 4566 |
| COPPER HYDROXIDE | 30 | 42 | 30 | 3502 | 3276 | 6668 |
| METRIBUZIN | 31 | 51 | 59 | 3462 | 2628 | 2007 |
| CHLORANTRANILIPROLE | 32 | 45 | 48 | 3219 | 3051 | 2674 |
| IMIDACLOPRID | 33 | 40 | 45 | 3148 | 3651 | 3028 |
| THIOPHANATE-METHYL | 34 | 36 | 43 | 2922 | 4007 | 3075 |
| ORYZALIN | 35 | 27 | 22 | 2766 | 6908 | 10620 |
| ETHEPHON | 36 | 38 | 46 | 2716 | 3751 | 2805 |
| TRIFLOXYSTROBIN | 37 | 54 | 60 | 2530 | 2404 | 1981 |
| SPINETORAM | 38 | 48 | 47 | 2434 | 3000 | 2794 |
| PYRACLOSTROBIN | 39 | 55 | 51 | 2417 | 2385 | 2613 |
| FLUMIOXAZIN | 40 | 85 | 105 | 2300 | 799 | 514 |
| SPINOSYN | 41 | 57 | 65 | 2293 | 1867 | 1523 |
| CYPRODINIL | 42 | 63 | 81 | 2244 | 1414 | 996 |
| PENTHIOPYRAD | 43 | 41 | 52 | 2140 | 3391 | 2593 |
| ACETAMIPRID | 44 | 53 | 58 | 2093 | 2406 | 2073 |
| SPIROTETRAMAT | 45 | 65 | 68 | 2042 | 1394 | 1388 |
| FLUOPYRAM | 46 | 68 | 84 | 2034 | 1313 | 901 |
| OXYFLUORFEN | 47 | 56 | 42 | 2031 | 2351 | 3136 |
| SETHOXYDIM | 48 | 90 | 103 | 1912 | 695 | 548 |
| PROHEXADIONE | 49 | 61 | 71 | 1869 | 1418 | 1313 |
| METHOXYFENOZIDE | 50 | 60 | 66 | 1736 | 1490 | 1511 |
| MYCLOBUTANIL | 51 | 64 | 72 | 1732 | 1414 | 1267 |
| DIMETHOATE | 52 | 73 | 73 | 1616 | 1125 | 1234 |
| CUPROUS OXIDE | 53 | 35 | 25 | 1537 | 4064 | 9271 |
| OXYTETRACYCLINE | 54 | 58 | 64 | 1491 | 1598 | 1561 |
| DICAMBA | 55 | 83 | 85 | 1400 | 817 | 897 |
| AUREOBASIDIUM PULLULANS | 56 | 95 | 99 | 1378 | 591 | 591 |
| CHLOROPICRIN | 57 | 49 | 35 | 1298 | 2904 | 4925 |
| FENPROPATHRIN | 58 | 71 | 86 | 1271 | 1285 | 819 |
| BACILLUS SUBTILIS | 59 | 87 | 78 | 1242 | 746 | 1106 |
| INDAZIFLAM | 60 | 79 | 92 | 1138 | 945 | 695 |
| NOVALURON | 61 | 80 | 88 | 1127 | 943 | 799 |
| BROMOXYNIL | 62 | 37 | 49 | 1077 | 3878 | 2657 |
| 2,4-DB | 63 | 96 | 100 | 1065 | 588 | 588 |
| RIMSULFURON | 64 | 78 | 93 | 1049 | 1009 | 689 |
| QUINOXYFEN | 65 | 84 | 91 | 889 | 806 | 703 |
| MCPA | 66 | 76 | 87 | 877 | 1021 | 809 |
| NORFLURAZON | 67 | 59 | 61 | 832 | 1525 | 1715 |
| DIMETHENAMID & DIMETHENAMID-P | 68 | 69 | 77 | 803 | 1287 | 1139 |
| DIMETHENAMID-P | 69 | 70 | 82 | 803 | 1287 | 968 |
| TRI-ALLATE | 70 | 106 | 109 | 799 | 378 | 438 |
| DIFENOCONAZOLE | 71 | 98 | 114 | 790 | 507 | 364 |
| SIMAZINE | 72 | 30 | 37 | 767 | 4873 | 4527 |
| FLUTRIAFOL | 73 | 88 | 98 | 766 | 736 | 596 |
| HEXYTHIAZOX | 74 | 91 | 104 | 743 | 691 | 520 |
| TEBUCONAZOLE | 75 | 94 | 110 | 735 | 612 | 405 |
| BIFENAZATE | 76 | 74 | 76 | 732 | 1069 | 1160 |
| HEXAZINONE | 77 | 67 | 75 | 659 | 1322 | 1168 |
| FLUBENDIAMIDE | 78 | 121 | 141 | 629 | 261 | 217 |
| PYRIPROXYFEN | 79 | 93 | 102 | 590 | 618 | 584 |
| DIAZINON | 80 | 46 | 50 | 549 | 3040 | 2653 |

# Appendices

Table 2: Top 50 Estimates in Yakima County, Range (EPest-low and EPest-high), 1-Yr, 2016

| RANK_1YR | COMPOUND | EPEST_LOW_KG_1YR_ONLY | EPEST_HIGH_KG_1YR_ONLY |
|---|---|---|---|
| 1 | PETROLEUM OIL | 1715390 | 1715390 |
| 2 | SULFUR | 397464 | 397630 |
| 3 | CALCIUM POLYSULFIDE | 197137 | 197137 |
| 4 | KAOLIN CLAY | 151792 | 153162 |
| 5 | GLYPHOSATE | 89580 | 89873 |
| 6 | COPPER | 51880 | 51880 |
| 7 | CHLORPYRIFOS | 40276 | 40284 |
| 8 | COPPER SULFATE | 35194 | 35194 |
| 9 | PARAQUAT | 26038 | 26162 |
| 10 | METAM | 23265 | 23355 |
| 11 | DICHLOROPROPENE | 21475 | 21475 |
| 12 | 2,4-D | 17998 | 18582 |
| 13 | SULFURIC ACID | NA | 17168 |
| 14 | CARBARYL | 14114 | 14195 |
| 15 | PENDIMETHALIN | 11958 | 12959 |
| 16 | ZIRAM | 11924 | 11924 |
| 17 | METAM POTASSIUM | NA | 10763 |
| 18 | MANCOZEB | 10218 | 10218 |
| 19 | ATRAZINE | 61 | 7221 |
| 20 | TRIFLUMIZOLE | 7031 | 7031 |
| 21 | POTASSIUM BICARBONATE | 6593 | 6593 |
| 22 | BUPROFEZIN | 5292 | 5292 |
| 23 | GLUFOSINATE | 771 | 5018 |
| 24 | TRICLOPYR | 0 | 4962 |
| 25 | BOSCALID | 4376 | 4376 |
| 26 | BACILLUS THURINGIENSIS | 4314 | 4314 |
| 27 | PHOSMET | 3909 | 3909 |
| 28 | MALATHION | 3874 | 3888 |
| 29 | DIURON | 732 | 3519 |
| 30 | COPPER HYDROXIDE | 3392 | 3502 |
| 31 | METRIBUZIN | 3453 | 3462 |
| 32 | CHLORANTRANILIPROLE | 3211 | 3219 |
| 33 | IMIDACLOPRID | 3145 | 3148 |
| 34 | THIOPHANATE-METHYL | 2802 | 2922 |
| 35 | ORYZALIN | 2766 | 2766 |
| 36 | ETHEPHON | 2716 | 2716 |
| 37 | TRIFLOXYSTROBIN | 2530 | 2530 |
| 38 | SPINETORAM | 2434 | 2434 |
| 39 | PYRACLOSTROBIN | 2224 | 2417 |
| 40 | FLUMIOXAZIN | 2297 | 2300 |
| 41 | SPINOSYN | 2290 | 2293 |
| 42 | CYPRODINIL | 2244 | 2244 |
| 43 | PENTHIOPYRAD | 2105 | 2140 |
| 44 | ACETAMIPRID | 2093 | 2093 |
| 45 | SPIROTETRAMAT | 2021 | 2042 |
| 46 | FLUOPYRAM | 2034 | 2034 |
| 47 | OXYFLUORFEN | 1900 | 2031 |
| 48 | SETHOXYDIM | 1912 | 1912 |
| 49 | PROHEXADIONE | 1869 | 1869 |
| 50 | METHOXYFENOZIDE | 1685 | 1736 |

Table 3: Top 50 Estimates in Yakima County, Range (EPest-low and EPest-high), 3-Yr Avg, 2014-2016

| RANK_3YR | COMPOUND | EPEST_LOW_KG_3YR_AVG | EPEST_HIGH_KG_3YR_AVG |
|---|---|---|---|
| 1 | PETROLEUM OIL | 1523056 | 1523056 |
| 2 | SULFUR | 339241 | 341576 |
| 3 | CALCIUM POLYSULFIDE | 199218 | 199437 |
| 4 | KAOLIN CLAY | 126292 | 127777 |
| 5 | CAPTAN | 125 | 98350 |
| 6 | GLYPHOSATE | 63789 | 64993 |
| 7 | CHLORPYRIFOS | 43076 | 43552 |
| 8 | COPPER | 38076 | 38083 |
| 9 | DICHLOROPROPENE | 36107 | 36697 |
| 10 | METAM | 20438 | 33961 |
| 11 | CARBARYL | 24448 | 24652 |
| 12 | 2,4-D | 15096 | 21719 |
| 13 | COPPER SULFATE | 18059 | 19198 |
| 14 | PINOLENE | NA | 18606 |
| 15 | DODINE | NA | 18532 |
| 16 | PARAQUAT | 15778 | 16855 |
| 17 | SULFURIC ACID | NA | 15226 |
| 18 | METAM POTASSIUM | NA | 15213 |
| 19 | MANCOZEB | 13729 | 14701 |
| 20 | PENDIMETHALIN | 13530 | 14692 |
| 21 | ZIRAM | 11906 | 11906 |
| 22 | TRIFLUMIZOLE | 8596 | 8596 |
| 23 | ACEPHATE | NA | 8584 |
| 24 | METOLACHLOR & METOLACHLOR-S | 154 | 8162 |
| 25 | METOLACHLOR-S | 154 | 8159 |
| 26 | POTASSIUM BICARBONATE | 7685 | 7877 |
| 27 | ORYZALIN | 5868 | 6908 |
| 28 | PHOSMET | 6154 | 6154 |
| 29 | DIURON | 2519 | 5254 |
| 30 | SIMAZINE | 4873 | 4873 |
| 31 | HYDROGEN PEROXIDE | 685 | 4615 |
| 32 | BOSCALID | 4442 | 4467 |
| 33 | BACILLUS AMYLOLIQUEFACIEN | NA | 4446 |
| 34 | GLUFOSINATE | NA | 4269 |
| 35 | CUPROUS OXIDE | 3887 | 4064 |
| 36 | THIOPHANATE-METHYL | 1974 | 4007 |
| 37 | BROMOXYNIL | 903 | 3878 |
| 38 | ETHEPHON | 3751 | 3751 |
| 39 | ATRAZINE | 56 | 3660 |
| 40 | IMIDACLOPRID | 3586 | 3651 |
| 41 | PENTHIOPYRAD | 3370 | 3391 |
| 42 | COPPER HYDROXIDE | 3205 | 3276 |
| 43 | BUPROFEZIN | 3267 | 3267 |
| 44 | TRICLOPYR | NA | 3118 |
| 45 | CHLORANTRANILIPROLE | 3029 | 3051 |
| 46 | DIAZINON | 2997 | 3040 |
| 47 | BACILLUS THURINGIENSIS | 3014 | 3014 |
| 48 | SPINETORAM | 2987 | 3000 |
| 49 | CHLOROPICRIN | 2500 | 2904 |
| 50 | MALATHION | 2687 | 2710 |

Table 4: Top 50 Estimates in Yakima County, Range (EPest-low and EPest-high), 5-Yr Avg, 2012-2016

| RANK_5YR | COMPOUND | EPEST_LOW_KG_5YR_AVG | EPEST_HIGH_KG_5YR_AVG |
|---:|---|---:|---:|
| 1 | PETROLEUM OIL | 1431482 | 1431482 |
| 2 | SULFUR | 283777 | 285379 |
| 3 | CALCIUM POLYSULFIDE | 162911 | 164121 |
| 4 | KAOLIN CLAY | 129623 | 131143 |
| 5 | CAPTAN | 262 | 59808 |
| 6 | GLYPHOSATE | 55518 | 56610 |
| 7 | DICHLOROPROPENE | 38931 | 41146 |
| 8 | METAM | 28773 | 40436 |
| 9 | CHLORPYRIFOS | 39726 | 40193 |
| 10 | METAM POTASSIUM | NA | 29218 |
| 11 | COPPER | 28061 | 28067 |
| 12 | CARBARYL | 21766 | 21888 |
| 13 | 2,4-D | 13121 | 21304 |
| 14 | SULFURIC ACID | NA | 18163 |
| 15 | PARAQUAT | 15298 | 16586 |
| 16 | PENDIMETHALIN | 14565 | 15812 |
| 17 | COPPER SULFATE | 13970 | 15423 |
| 18 | TRICLOPYR | NA | 14269 |
| 19 | MANCOZEB | 12718 | 13483 |
| 20 | THIRAM | NA | 12539 |
| 21 | DODINE | NA | 12147 |
| 22 | ORYZALIN | 9970 | 10620 |
| 23 | PINOLENE | NA | 9903 |
| 24 | ZIRAM | 9800 | 9892 |
| 25 | CUPROUS OXIDE | 2607 | 9271 |
| 26 | GLUFOSINATE | NA | 8831 |
| 27 | ACEPHATE | NA | 8584 |
| 28 | TRIFLUMIZOLE | 7915 | 7915 |
| 29 | POTASSIUM BICARBONATE | 6826 | 6942 |
| 30 | COPPER HYDROXIDE | 6343 | 6668 |
| 31 | METOLACHLOR & METOLACHLOR-S | 146 | 6369 |
| 32 | ENDOSULFAN | 1523 | 6354 |
| 33 | PHOSMET | 5792 | 5792 |
| 34 | METOLACHLOR-S | NA | 5723 |
| 35 | CHLOROPICRIN | 3788 | 4925 |
| 36 | DIURON | 2423 | 4566 |
| 37 | SIMAZINE | 4507 | 4527 |
| 38 | BACILLUS AMYLOLIQUEFACIEN | NA | 4446 |
| 39 | BOSCALID | 4058 | 4099 |
| 40 | ATRAZINE | 1894 | 4056 |
| 41 | AZINPHOS-METHYL | 3619 | 3619 |
| 42 | OXYFLUORFEN | 2872 | 3136 |
| 43 | THIOPHANATE-METHYL | 1773 | 3075 |
| 44 | HYDROGEN PEROXIDE | 528 | 3067 |
| 45 | IMIDACLOPRID | 2981 | 3028 |
| 46 | ETHEPHON | 2805 | 2805 |
| 47 | SPINETORAM | 2787 | 2794 |
| 48 | CHLORANTRANILIPROLE | 2660 | 2674 |
| 49 | BROMOXYNIL | 809 | 2657 |
| 50 | DIAZINON | 2625 | 2653 |